Graph Theory

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1 Basics

Social Network: This is a type of network that shows different people and their relationships or interactions with each other.

Vertex: In a graph, the vertices or nodes are the objects being connected. These are the elements of the graph.

Edge: The edges are the connections between nodes. They represent the relationships or connections between different elements within the graph.

2 Measures of Centrality

Centrality: This is a measure of how important a node is in the graph. The centrality of a node can be measured in many different ways, depending on the type of graph and connections.

Degree: The degree of a node is how many edges it has connected to it. When using degree to measure centrality, more edges would indicate more importance for each vertex.

Closeness: Closeness measures how a node is to the other nodes in the network by averaging the shortest paths to all other nodes. The closer a node is to other nodes, the more central it is.

Betweeness: This measures the number of times a certain vertex appears in the shortest paths between other nodes. We can use this to measure centrality by finding the shortest paths between all of the vertices and counting which node appears the most times.

Eigenvector: The eigenvector centrality of a node is calculated by considering the vertex and its neighbors' connections. We can do this by finding the nonnegative eigenvector of the graph's adjacency matrix and taking one of its components.

Example: Consider that a Girls Scout group is attempting to market their cookies in person to increase sales. Since there is a limited amount of houses they can all go to, they want to find the most central person in their social network to target. Consider the adjacency matrix A = $\begin{bmatrix} & v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \\ v_6 & 0 & 1 & 0 & 1 & 0 & 0 \\ v_3 & 1 & 0 & 1 & 0 & 0 & 1 \\ v_4 & 0 & 1 & 0 & 1 & 0 & 0 \\ v_5 & 1 & 0 & 1 & 0 & 1 & 0 \\ v_1 & 0 & 0 & 0 & 1 & 0 & 1 \\ v_2 & 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$. Using all measures of centrality, which person should the Girl Scouts market to?